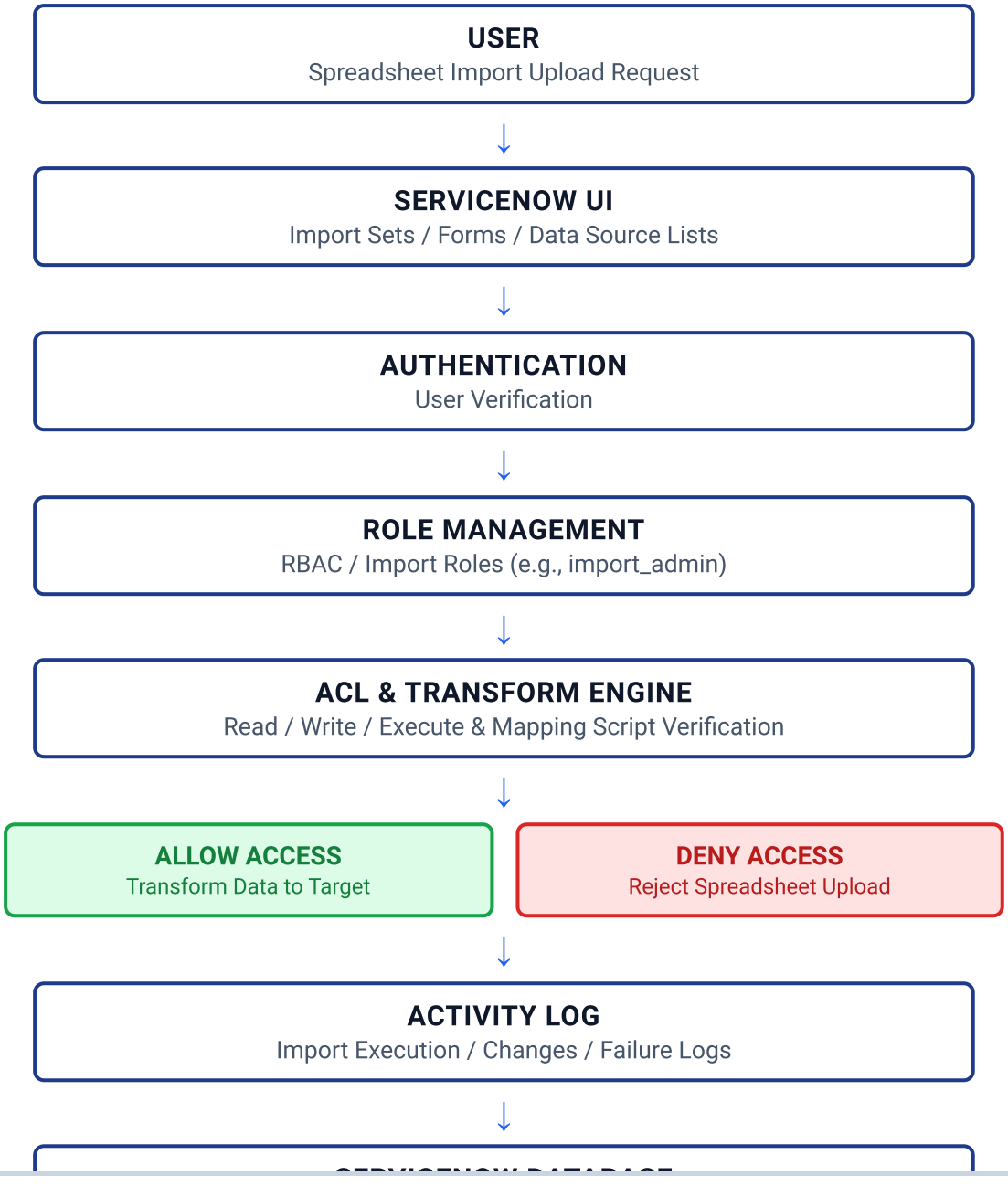
TECHNICAL ARCHITECTURE DOCUMENT

Import Data Using Transform Maps

Solution Architecture diagram, execution pipeline flow, and component specifications for importing spreadsheet data using transform maps.

3. PROJECT DESIGN PHASE: SOLUTION ARCHITECTURE

The Solution Architecture defines the end-to-end pipeline through which spreadsheet records flow—from file submission at the UI layer down to database modification, governed by authentication, RBAC, ACL rules, and the Transform Map engine.



4. ARCHITECTURE COMPONENTS & FLOW

User Interface Layer

- Users interact via ServiceNow Data Sources, Import Sets, Forms, and Lists.
- Allows uploading `.xlsx`, `.xls`, and `.csv` spreadsheets.

Authentication Layer

- Verifies identity and credentials before staging files.
- Ensures external users cannot access staging queues.

Role Management (RBAC)

- Determines user privileges using roles (`import_admin`, `import_transformer`).
- Enforces separation of duties between upload and mapping.

ACL & Transform Engine

- Checks Read, Write, and Execute permissions on tables and scripts.
- Maps staging columns to target fields with Coalesce key rules.

Activity & Transform Logging

- Records every import attempt, row transformation, and error.
- Provides real-time execution logs for auditing.

ServiceNow Database

- Stores Users, Roles, ACL Rules, Staging Tables, Transform Maps, and Target Records.

ARCHITECTURE FLOW SUMMARY

Execution Pipeline Order:

User → ServiceNow UI → Authentication → Role Validation → ACL & Transform Check → Allow/Deny Decision → Activity Log → ServiceNow Database